

STATEMENT OF WORK

Repair Catwalks, B3001: Post S-57

Project #: WWYK200326

Work Task #: 23951

Revision: 0

Issued: 08 JAN 26

1. SCOPE

1.1 GENERAL

The work to be performed under the terms of this Order consists of furnishing all design/engineering services, materials, labor, plant, tools, transportation, equipment, testing services, quality control and incidentals required to accomplish the Installation of FOD Guards on Catwalks above F135 Shop, Post S57 at Tinker Air Force Base (AFB), Oklahoma (reference the project drawings for additional location information).

The Contractor shall perform a site investigation to verify existing conditions. If necessary, additional as-built drawings are available from 72 ABW/CE - Engineering Support, located in B400.

The Contractor shall be responsible for identifying the necessary logistical, architectural, and structural work elements required to perform the work as described herein. Assemblies, framework, fixtures, or apparatus not readily identified on the project drawings or this Statement of Work (SOW) shall be provided by the Contractor as required for a complete, fully operating, and functioning system or systems ready for use by the Government.

Information such as floor plan layouts and dimensions; structural, mechanical, and electrical data; and location of underground utilities provided by the Government for existing and new construction are based on available as-built documentation, field observation, and equipment selected as design basis by the Government. This information is to be considered approximate and shall be field verified by the Contractor.

1.2 DEFINITIONS

1.2.1 Contracting Officer (CO)

An individual with the authority to enter into, administer, and/or terminate contracts and make related determinations and findings. The term includes certain authorized representatives of the contracting officer acting within the limits of their authority as delegated by the CO.

The CO is the ONLY person authorized to make any commitments or changes that will affect price, quality, quantity, delivery, or any other term or condition of the contract.

1.2.2 Contracting Officer's Representative (COR)

An individual designated by the Contracting Officer to perform specific technical or administrative functions.

The COR is not authorized to make any commitments or changes that will affect price, quality, quantity, delivery, or any other term or condition of the Order. It shall be the Contractor's responsibility to contact the CO concerning any matters where the Contractor believes that the

direction received from the COR may affect price, quality, quantity, delivery, or any other term or condition of the Order prior to performing work or purchase of material.

1.3 CONTRACTOR PERSONNEL

1.3.1 General Requirements

The Contractor shall ensure that employees and subcontractor employees have current/valid licenses and certifications necessary to perform their functions. Any cost or time associated with obtaining licenses or certifications shall be at the Contractor's expense.

Personnel shall be easily recognized as a Contractor employee or subcontractor. This may be accomplished by wearing distinctive clothing bearing the name of the company or by wearing appropriate badges that contain the company name and employee name. The badge shall be displayed on the outer clothing, on the front of the body between the neck and waist and must be visible at all times. The badge shall clearly identify the individual as a Contractor employee.

1.3.2 Key Personnel

The Contractor shall, at a minimum, provide the following key personnel. Submit resumes for personnel fulfilling each of the key positions described below within five (5) calendar days after NTP.

Exceptions or variances to experience requirements must have prior approval from the Contracting Officer (CO)/Contracting Officer's Representative (COR); exceptions/variances will be considered where the Contractor can demonstrate comparable experience and/or competency.

Forward any requests for variance to the CO; include resumes for personnel listing the qualifications to be considered (e.g. education, experience, professional certifications, etc.).

Required experience levels shall be maintained throughout the duration of the agreement. When personnel changes are required, resumes and a revised organization chart shall be submitted to the CO within five (5) business days of a vacated or filled position.

One Contractor employee may hold up to two (2) key positions; exceptions are noted below (e.g. a Project Superintendent cannot fulfill the role of the Quality Control Manager). Contractor employees holding more than one key position must meet the minimum requirements for both positions.

1.3.2.1 Project Manager (PM)

The PM will be responsible for the overall performance of this Order. The PM shall have a minimum of ten (10) years combined experience in the following positions: Project Superintendent, Quality Control Manager, Project Manager, Project Engineer, or Construction Manager with general contracting firm(s) engaged in multi-trade commercial construction contracts. The individual must have at least two (2) years' experience as a Project/Program Manager. Provide the name of PM, in writing, to the CO upon award of the Order.

The Contractor shall designate an Alternate PM who may act on behalf of the Contractor when the PM is absent. Qualifications for the Alternate are the same as for the PM. Provide the name of the Alternate, in writing, to the CO upon award of the Order.

The PM or Alternate shall have direct oversight of project planning, design, and construction activities. The PM shall provide overall Order management, including full authority to develop proposals, negotiate, and sign modifications to Orders; provide subcontract administration; review material submittals and shop drawings; and shall attend all status meetings, site visits and pre-final/final inspections.

The CO must be able to contact the PM or Alternate by phone at all times. Throughout the life of this agreement, the PM or Alternate shall be available to meet with the CO at Tinker AFB within one (1) hour of telephone notification during normal duty hours.

The PM or the alternate may fulfill the role of the Design Project Manager.

1.3.2.2 Design Project Manager

The Design Project Manager will be responsible for the overall performance of the pre-construction/design portions of this contract. The Design Project Manager shall be licensed in the State of Oklahoma as a Registered Architect or Professional Engineer; Design Project Manager shall have a minimum of ten (10) years' previous experience in project management, with at least five (5) of those years managing design-build projects similar in size and complexity. The name of the Design Project Manager shall be designated in writing to the CO upon award of the Order.

The Design Project Manager shall provide direct oversight and overall management of the design process, including coordinating day to day production; schedule adherence; quality control of all design submittals; verification that designs meet applicable codes and standards (including Air Force/DoD standards); review of material submittals and shop drawings; and answering construction RFIs related the design. The Design Program Manager shall attend all design status meetings, initial pre-design meetings, site visits, and design review conferences.

The Design Project Manager may fulfill the role of the Project Manager or alternate.

1.3.2.3 Quality Control Manager (QCM)

The QCM is responsible for management of the Contractor Quality Control (CQC) Program. The QCM shall have the minimum qualifications set forth in specification section 01 45 00.00 10, *Quality Control* (for purposes of this SOW, QCM is equivalent to CQC System Manager).

The Contractor shall designate an Alternate QCM who may act on behalf of the Contractor when the QCM is absent. Qualifications for the Alternate(s) are the same as for the QCM.

Contractor may designate additional quality control (QC) staff as specified in 01 45 00.00 10 (e.g. Assistant QC Manager or QC specialists).

Provide the names of QC staff to the CO upon award of the Order as a part of the Quality Control Plan.

The QCM and their staff shall coordinate and review all material and shop drawing submittals, including those of subcontractors, offsite fabricators, and suppliers; prepare quality control (QC) reports; schedule and coordinate testing procedures; and attend all status meetings, site visits, and pre-final/final inspections.

The Contractor's QC staff shall maintain a presence at the site at all times when work is being performed to ensure that the materials and workmanship are in accordance with the project drawings and technical specifications, as well as approved shop drawings and material submittals. The QCM or Alternate shall visit the project site daily.

For additional information regarding the requirements for Contractor's Quality Control Program, refer to specification section 01 45 00.00 10.

The QCM may serve as the Site Safety and Health Officer.

1.3.2.4 Site Safety and Health Officer (SSHO)

The SSHO is responsible for management of the Contractor Health and Safety Program (for purposes of this SOW, the Health and Safety Program is equivalent to Safety and Occupational Health Program in EM 385-1-1, *U.S. Army Corps of Engineers Safety and Health Requirements Manual*). The SSHO shall have the minimum qualifications set forth in EM 385-1-1.

The Contractor shall designate an Alternate SSHO who may act on behalf of the Contractor when the SSHO is absent. Qualifications for the Alternate(s) are the same as for the SSHO.

Provide the names of the SSHO and Alternate to the CO upon award of the Order as a part of the Health and Safety Plan.

The SSHO or Alternate shall visit the construction site daily to ensure full compliance with all health and safety requirements.

When the SSHO or Alternate is not on site, a Designated Representative (DR) may be used (reference EM 385-1-1). The SSHO, an Alternate, or a DR shall be on the construction site at all times when work is being performed.

1.3.2.5 Project Superintendent

The Superintendent must have a minimum of five (5) years' experience in construction with at least two (2) of those years as a superintendent on commercial projects similar in size and complexity to the requirements of this Order. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance.

The Contractor shall designate an Alternate Superintendent who may act on behalf of the Contractor when the Superintendent is absent. Qualifications for the Alternate are the same as for the Superintendent, except previous experience as a superintendent is not required; subcontractors will not be used as Alternates.

Provide the names of the Superintendent and the Alternate to the CO upon award of the Order.

The Superintendent is primarily responsible for managing and coordinating day-to-day production and schedule adherence for the Order. Superintendents shall provide on-site supervision; submit progress reports; schedule and coordinate subcontractors and material suppliers; and attend all status meetings, site visits and pre-final/final inspections.

An individual Superintendent may be assigned to more than one Order; however, the Superintendent shall not be assigned to more than 5 active Orders at Tinker AFB unless approved by the CO. Alternates may only be assigned to a single Order/job site.

The Superintendent shall visit the Order/job site daily. Either the Superintendent or the Alternate shall be at the job site at all times when work is being performed. The Superintendent must arrive at the job site within 30 minutes of telephone direction from the CO, Contract Specialist, or Construction Quality Assurance (QA) Inspector.

The Superintendent or Alternate(s) cannot be a member of the QC staff.

The Superintendent or Alternate(s) can be the Designated Representative (reference section SITE SAFETY AND HEALTH OFFICER elsewhere in this document).

1.3.3 Subcontractors

Subcontractor firms performing design shall be authorized to offer engineering or architectural services in a US State or Territory. An Oklahoma license will not be required for individuals responsible for signing/sealing documents.

All subcontractor firms shall be licensed in their specialty where required; Contractor will provide a copy of these licenses as part of the QC Plan.

1.4 BADGES

AFMC Form 387, *Air Force Material Command Identification Cards* (i.e. contractor badges) are required for Contractor personnel to access/work on Tinker AFB. The Contractor must have all personnel badged prior to commencing work.

Contractor shall designate an employee to act as Certifying Official for all personnel requiring badges. The designated Certifying Official shall complete the *Appointment of Security Representative* letter and submit to the COR upon award of the Order; a copy of this letter (Attachment 1) is included on the attached CD.

Upon approval, Contractor's Certifying Official will be responsible for completing AFMC Form 496, *Application for AFMC Identification Card*, for all personnel requiring a contractor badge. Badges are issued at Bldg. 6611, Pass and ID.

Badges shall be displayed on the outer clothing, on the front of the body between the neck and waist and must be visible at all times.

1.5 QUALITY CONTROL

It is the Contractor's responsibility to provide, inspect, and document the level of quality that has been established by the minimum standards of the project drawings, specifications, and this SOW. The QCM shall direct the execution of the Contractor's QC Plan.

Quality control shall be executed in accordance with specification section 01 45 00.00 10. The QC program shall include, but is not limited to the following:

1.5.1 Inspection System

The Contractor shall establish an inspection system which monitors the execution of any task completed under this Order. The Contractor must specify the areas to be inspected on a scheduled or unscheduled basis, how often inspections will be accomplished, and titles of the individuals who will perform the inspections.

1.5.2 Deficiency Prevention

The Contractor shall establish a method for identifying, correcting, and preventing future recurrence of quality deficiencies.

1.5.3 Inspection Files

In accordance with specification section 01 45 00.00 10, the Contractor shall maintain records of all required inspections conducted by the QCM and the necessary corrective actions taken. This documentation shall be submitted to the Government weekly with the progress reports (reference section PROGRESS REPORTS elsewhere in this document) throughout the life of the Order.

1.6 HEALTH AND SAFETY

1.6.1 Commander's Statement of Safety Intent

The safety of everyone on Tinker AFB is our top priority. All individuals must accept personal responsibility for safety and seek opportunities for improving the safety of our work environment. Everyone working on Tinker AFB is challenged to find, fix (if able) and report all hazards and near-misses. It is our shared duty to take every reasonable step and precaution to prevent accidents and preserve the health/safety of each contractor and government employee.

1.6.2 Occupational Safety and Health Administration (OSHA) Voluntary Protection Program (VPP)

The Contractor is hereby notified that TAFB has achieved or is pursuing VPP Star ratings across multiple sites. Additional information regarding VPP can be found at the following link:

<http://www.osha.gov/dcsp/vpp/index.html>

The Voluntary Protection Plan applies to all construction and service contracts that have Contractor employees working more than 1000 hours per quarter on Tinker AFB. Refer to specification section 00 71 00, *General Safety Requirements for Construction on Tinker AFB* for additional requirements related to VPP.

1.6.3 Emergency Evacuation Procedures

Contractor employees shall follow the direction of Government employees in regard to emergency evacuation procedures, i.e. fire, tornado, or bomb threats. It is the responsibility of the Contractor to provide guidance and establish procedures for responding to an emergency for their employees. It is the Contractor's responsibility to account for their employees following an emergency evacuation. The Contractor personnel shall participate with the Government personnel in all fire and tornado drills at no additional cost to the Government.

1.6.4 Mishap Reporting

The Contractor shall notify the CO and the COR as soon as practical after any mishaps, including recordable accidents, incidents, and near misses as defined in EM 385-1-1 Appendix Q; and any injuries, illness, or property damage which occur on the installation while performing work under this Order.

Refer to specification section 00 71 00 for additional requirements related to mishap reporting.

1.7 BIOENVIRONMENTAL REGULATIONS

- (1) To assist in the evaluation of health hazards & to provide emergency medical treatment, Contractor must have Material Data Sheets (MDS) for hazardous materials reviewed and

exchange of HAZCOM information accomplished prior to use. Process hazardous material in accordance with specification section 00 72 00, *Environmental Requirements for Construction on Tinker AFB*.

- (2) Hazardous noise levels in the vicinity of DoD Active Duty and Civilian personnel must be in accordance with OSHA 29 CFR 1910.95 and AFI 48-127, *Occupational Noise and Hearing Conservation Program*.

1.8 OUTAGES AND CLOSURES

Request utility outages and/or closures in accordance with TAFB Standard Operating Procedure *Planned and Emergency Outages*. Outages may be required to be performed outside of normal duty hours. The Contractor shall provide temporary lighting and power that may be required for continued performance of User's mission during outages.

1.9 EXISTING WORK

In accordance with FAR 52.236-9, *Protection of Existing Vegetation, Structures, Equipment, Utilities, and Improvements*, the Contractor shall take necessary precautions to avoid damage to existing items to remain in place, to be reused, or to remain the property of the Government; any damaged items shall be repaired or replaced as directed by the CO.

The Contractor shall remove or alter existing work in such a manner as to prevent injury or damage to any portions of the existing work which remain.

The Contractor shall repair or replace portions of existing work which have been altered during construction operations to match existing or adjoining work, as approved by the CO. At the completion of operations, existing work must be in a condition equal to or better than that which existed before new work started.

2. PROJECT SPECIFIC REQUIREMENTS

2.1 GOVERNMENT OCCUPANCY OF FACILITIES

The facility will remain occupied throughout construction.

The construction area, a reasonable buffer area, and any approved staging areas will be cleared of all Government materials and personnel prior to the start of construction.

Contractor shall consult with the Government Project Manager/Construction QA Inspector and the Using Agency to coordinate procedures; means of access; space for storage of materials and equipment; and use of approaches, corridors, and stairways in such a way that accommodates the continued performance of the User's mission during construction.

2.2 WORKING HOURS

In accordance with specification section 00 70 00, *General Requirements for Construction on Tinker Air Force Base*, construction shall take place during normal duty hours (0700 – 1600 Monday-Friday); utility outages will be the exception.

The Contractor has the responsibility to manage and maintain the construction schedule; the Contractor may request authorization to work weekends, holidays, and additional shifts as required to accomplish the work per their baseline schedule (reference section CONSTRUCTION SCHEDULE elsewhere in this document).

Work outside normal duty hours to meet schedule will be performed at no additional cost to the Government; this is to include all testing, inspection, and acceptance actions.

2.3 PHASING

There is no Government-required construction phasing for this Order.

2.4 FOD AWARENESS AND PREVENTION TRAINING

In accordance with specification section 00 71 00 all Contractor personnel working on this Order will be required to complete Foreign Object Debris (FOD) awareness and prevention training.

The work to be performed for this Order will require the Contractor to work in or travel through areas near operational and/or production aircraft or related components.

2.5 AIRFIELD OPERATIONS

The work to be performed under this Order is not located in an Airfield Operations area.

2.6 CONTROLLED/RESTRICTED AREA(S) – B3001

Work to be performed under this Order is located within the controlled area of Bldg. 3001 (the Industrial Floor); a Bldg. 3001 Visitor Badge is required for Contractor personnel to access the work site.

Contractor shall designate an employee(s) to act as escort for all personnel requiring access to the controlled area/work site; subcontractors will not be used as escorts.

Note: the minimum requirement is one (1) escort for every ten (10) Contractor personnel; Contractor should anticipate providing at least two (2) escorts for every ten (10) personnel.

The designated escorts shall complete Bldg. 3001 controlled area training; training slides and video (Attachments 2a/b) are included on the attached CD. Document completion by signing the certificate (Attachment 2c) included in the training materials and submit to the Bldg. 3001 Visitor Center, Post AB-76 (i.e. Hollywood and Vine); alternatively, an electronic copy of the completed form can be emailed to the following address prior to obtaining badges:

OC-Complex.ContingencyCell@us.af.mil

Upon receipt of the visitor badge(s), the Contractor will be required to sign an AF Form 1297, *Temporary Issue Receipt*. Submit a copy of the signed badge issue receipt to the COR or their designated representative (i.e. Construction QA Inspector).

Upon completion of the work, return the visitor badge(s) to the Bldg. 3001 Visitor Center. Submit a copy of the signed badge return receipt to the COR or their designated representative as part of closeout documentation.

2.7 AFFIRMATIVE PROCUREMENT

The Government has performed an initial assessment and determined that no products listed in the EPA Comprehensive Procurement Guideline, USDA Biopreferred, Energy Star, or FEMP Low Standby Power programs are anticipated to be used in this Order.

A blank version of the Recovered Materials Determination Form (RMDF) has been provided as an attachment to this SOW (Attachment 3). Throughout the design process, the Contractor shall review the included RMDF and add products if/when applicability is determined.

2.8 ENVIRONMENTAL MANAGEMENT (EM) SURVEY

An EM survey indicates there are no asbestos containing materials (ACM), heavy metals (HM), or lead based paint (LBP) concerns for area(s) where work is to be performed. If at any time the Contractor encounters suspect environmental conditions, they shall notify the CO and the COR.

If grinding or disturbance occurs on painted or flat surfaces, then testing of Asbestos/Chromium presence will be required prior to work.

If at any time the Contractor encounters suspect environmental conditions not identified above, they shall notify the CO and the COR.

Follow all applicable USAF, OSHA, and UFC Occupational Health and Requirements.

3. PRECONSTRUCTION & CONSTRUCTION SERVICES BY CONTRACTOR

3.1 CORRESPONDENCE

All electronic correspondence related to the awarded Order shall be addressed to the CO, Contract Specialist, Government Project Manager, Construction QA Inspector, the COR, and the respective organizational workflow accounts. Hardcopy correspondence will be addressed to the CO, the Contract Specialist, and the COR.

Contact information for the above will be provided at the Post Award meeting.

All electronic correspondence from the Contractor regarding a specific Order shall follow a standardized subject line naming convention. Subject line for all electronic correspondence shall be as follows:

- WWYK##### Project Title: Subject of E-mail

3.2 DOCUMENTATION

All documents shall be submitted in accordance with specification section 01 33 00, *Submittal Requirements*. The Contractor shall submit all documents electronically via optical disk, e-mail, or a file transfer service such as DoD SAFE; except as modified by section DESIGN SUBMITTALS below.

All documents submitted by the Contractor requiring a signature shall be submitted to the CO and the COR with an electronic digital signature. All documents submitted by the Contractor requiring further action and signature by the Government shall be capable of being digitally signed and saved by the Government representative(s) as required.

3.2.1 Document Format

3.2.1.1 Specifications/Reports/Material Submittals

Specifications, design analyses, reports for calculations, material submittals, warranty information, and similar documents shall be letter sized (8-1/2" x 11"), submitted in Adobe Acrobat (.pdf) format. Black and white documents are preferred but color copies will be accepted if they are necessary to convey intent.

3.2.1.2 Drawings

All drawings produced by the Contractor shall be created utilizing Computer-Aided Design and Drafting (CADD) software. The drawing files shall be in AutoCAD (.dwg)

format and adhere to the drawing standards contained in Tri-Services A/E/C CADD Standard Release 6.0.

Drawing size shall be ANSI D (22" x 34"). Drawing scale shall be at least 1"=30' for civil/site drawings and 1/8"=1' for all other plan drawings; text and other non-scale annotation elements shall be sized to produce clear and legible prints at half-size. Contractor shall use English scales and dimensions.

Drawings submitted by the Contractor shall be .pdf format; in addition, final design drawings and as-built record drawings shall be submitted in .dwg format. Hard copy submittals, when required, shall be printed at half-size (11" x 17").

Reference specification sections 01 33 00; 01 33 16.00 10, *Design Data (Design After Award)*; and 01 78 00, *Closeout Submittals*, for additional requirements regarding developing and submitting design, shop, and as-built drawings.

3.2.2 File Naming Conventions

All documents submitted by the Contractor shall follow a standardized file naming convention as indicated below. File naming conventions shall be strictly adhered to and may only be modified by the Contracting Officer and/or the COR. File names shall have no spaces.

The following items in the below naming conventions are defined as followed:

- WWYK##### = Project number as defined by Civil Engineering
Example: WWYK201234
- YYYYMMDD = Effective date of document or report
- ZZZ = Contractor's Company Initials
- 001 = Submittal Number or Serial Report Number

(1) Proposals:

WWYK#####_Proposal#_ZZZ_Document Description
Example: WWYK201234_Proposal1_ACS_Cost Proposal

(2) RFIs:

WWYK#####_RFI_001

(3) Submittals:

WWYK#####_FORM#_001 (Brief Description of Submittal)
Example: WWYK201234_AF3000_004 CeramicTile

(4) Daily Reports:

WWYK#####_Daily Report_001_YYYYMMDD

(5) Progress Reports:

WWYK#####_PR_001_YYYYMMDD

(6) Other Miscellaneous Documents/Reports:

WWYK#####_Form Name, Number, Or Description_YYYYMMDD

3.3 DESIGN/CONSTRUCTION SCHEDULE

In accordance with FAR 52.236-15, *Schedules for Construction Contracts*, the Contractor shall submit a baseline design/construction schedule to the COR for approval within ten (10) calendar days after the NTP is issued. An approved schedule will be required prior to commencement of work.

Contractor's schedule must demonstrate a reasonable and realistic sequence of activities which represent all work through the Order performance period. The schedule shall consider the major work elements, as well as submittal reviews, mobilization, material delivery times, anticipated weather delays, and final inspection/closeout.

3.3.1 Schedule Format

The schedule shall be developed in accordance with specification section 01 32 16.00 20, *Small Project Construction Progress Schedules*, utilizing the Network Analysis Schedule (NAS)/Critical Path Method format.

The schedule shall generally match the schedule submitted with the Contractor's accepted proposal, with the following additional information included:

- (1) Order start (NTP) and end dates.
- (2) Additional detail/breakdown of major work elements, as necessary.
- (3) Cost Loading data entered for all work items. Cost loading must total to 100 percent of the value of the Order.

Any partial progress payments for materials, as approved by the CO, shall be entered separately on the schedule.

The schedule will reserve, as a minimum, the final 10% of an Order for final acceptance. No progress payments will be made against this final 10% of the Order until all closeout items have been submitted, all punch list items have been resolved, and final inspection/acceptance has been completed.

3.3.2 Schedule Updates

Contractor shall keep the schedule current, reflecting actual start and end dates for work elements, activity progress, and plan for completing the remaining work. The Contractor shall submit the updated schedule at monthly intervals, with the first progress report of the month. Reference specification section 01 32 16.00 20 for additional information.

In addition, the Contractor shall submit updated baseline design/construction schedules when directed by the CO or the COR. Examples of when a new baseline schedule may be required include instances where the period of performance has been revised through modification to the Order; or as part of a 'get well plan' for construction that is significantly behind schedule.

3.3.3 Coordination with Government Activities

In developing the schedule, the Contractor shall account for the following milestones and Government review times in addition to construction activities:

DELIVERY SCHEDULE & SUBMITTALS		
Item of Work & Deliverables	Government Review Time (calendar days)	Remarks
Post-Award Conference/Pre-design Conference/Notice to Proceed (NTP)	1 day	At Tinker AFB

DELIVERY SCHEDULE & SUBMITTALS		
Item of Work & Deliverables	Government Review Time (calendar days)	Remarks
Baseline Schedule and Schedule of Values/Draft Progress Report	14 days	Provide within 10 days of NTP
35% Design Submittal	14 days	Government comments will be provided to contractor prior to AER
35% Project presentation to AERP and review conference	7 days	AERP meetings are held each Thursday at Tinker AFB or by teleconference
65% Design Submittal	14 days	
65% review conference	7 days	At Tinker AFB or by teleconference; may be waived by Contracting Officer
95% Pre-final Design Submittal	14 days	Submit with AF3000
95% review conference	7 days	At Tinker AFB or by teleconference; may be waived by Contracting Officer
100% Final Submittal (Issued for Construction)	7 days	Initial submittal provided electronically; provide final documents as specified elsewhere in this SOW after approval.
Government Back Check	7 days	
SD-01 Preconstruction Submittals (except as noted below)	14 days	Provide all preconstruction submittals (except design) minimum of 30-days prior to design completion/ construction NTP
Health and Safety Plan	21 days	
Entry Access List / Vehicle Entry Access List	14 days	Must be approved prior to obtaining access to controlled/secure areas
Notice of Proposed Construction (FAA Form 7460-1)	60 days	Must be provided by 1st business day of month

DELIVERY SCHEDULE & SUBMITTALS		
Item of Work & Deliverables	Government Review Time (calendar days)	Remarks
Request for Contractor Staging Area	30 days	Must be approved prior to mobilization
Abatement Submittals (asbestos, lead based paint)	21 days	Provide in event paint is disturbed.
Material Submittals (SD-03, SD-04, SD-07, SD-08)	14 days	Provide material submittals for major components of the work with 95% design submittal
Pre-Construction Conference/NTP: Begin Mobilization	1 day	At Tinker AFB; may be waived by CO
Outages	21 days	
Construction Submittals (SD-06, SD-09)	14 days	
Final/Closeout Submittals (SD-10, SD-11)	14 days	Must be approved prior to final inspection.
Pre-final Inspection	7-day Minimum Notice	Contractor must have completed internal pre-final and resolved all punch list items from this inspection
Final Inspection	7-day Minimum Notice	All final/closeout submittals must be approved. Contractor must have resolved all punch list items from pre-final inspection

Note: Information provided in this table is provided to assist the Contractor in developing the construction schedule. Review times may vary depending on the complexity of the submittal.

3.4 PROGRESS REPORTING

Progress reports are used for computing payments, and for monitoring the progress against the Contractor's baseline schedule.

The Contractor shall develop a progress report utilizing the 'Reports' feature of their scheduling software. Contractor developed reports shall provide the same information as AF Form 3065, *Contract Progress Report*. The progress report shall utilize the same work elements and schedule of values/cost loading data developed for the design/construction schedule.

Contractor shall submit a draft version of the progress report to the CO and the COR for approval within ten (10) calendar days after the NTP is issued (with the Baseline Schedule).

After approval, the Contractor shall submit progress reports weekly, or as directed by the CO, throughout the life of the Order. For each progress report, the Contractor shall include the following:

- (1) Percentage of work completed for each work element during the reporting period.
- (2) Cumulative percentage of work completed for each work element to date.
- (3) Cumulative percentage of work completed for all work elements (percent of Order completion).
- (4) Scheduled cumulative percentage of work completed for all work elements, based on the baseline Order schedule (scheduled percent of Order completion).

Progress report's will be reviewed by the Construction QA Inspector and/or the Project Manager, who will in turn provide a recommendation to the COR to approve/disapprove the progress report; approval of the Contractor's progress report does not constitute acceptance of the work as being 100% in compliance with the Order requirements (i.e. project specifications, drawings, and this SOW).

In the event that the Contractor's reported progress deviates from the baseline schedule, the CO may elect to hold progress meetings.

3.5 DESIGN/ENGINEERING

3.5.1 Contractor's Design Requirements

The Government has provided a SOW (this document), project narrative, and drawings (including as-built information) stating current site conditions and project objectives/requirements for the Contractor's use in preparing a complete design, ready for construction.

Design is to be accomplished in accordance with specification section 01 33 16.00 10. The Contractor's design effort shall include the following submissions:

- Preliminary Design (35%)
- Intermediate Design (65%)
- Pre-Final Design (95%)
- Final Design/Issued for construction (100%)

The Preliminary Design submittal (35%) shall be presented to the Architectural/Engineering Review Board (AERB); meetings are scheduled for every Thursday, 0800-100. At this meeting, the contractor will present the design package, discussing such items as: project scope, design approach, identified code deficiencies, deviations from the contract documents, etc.

3.5.2 Minimum Design Guidance Documents

The Contractor, as a minimum, is required to comply with the current editions of each of the following publications for design and construction. The term 'most current' is understood to mean, "Most current as of the date the solicitation is issued."

Codes/standards/regulations referenced by these publications are included as a part of this SOW to the extent referenced. In case of conflict between the project documents (SOW and

drawings), Unified Facility Criteria (UFCs), governing codes/standards, and the Tinker AFB facility standards, the Contractor shall submit a request for information (RFI) to the Government for resolution.

3.5.2.1 General

- (1) UFC 1-200-01, DoD Building Code
- (2) UFC 1-200-02, High Performance and Sustainable Building Requirements
- (3) UFC 4-010-01, DoD Minimum Antiterrorism Standards for Buildings
- (4) Tinker AFB Facility Standard

3.5.2.2 Structural

- (1) UFC 3-301-01, Structural Engineering
- (2) ASCE/SEI-7 Design loads for Buildings and Other Structures
- (3) AISC Specification for Structural Steel Buildings

3.5.3 Design Submittals

All design submittals shall be submitted in accordance with specification sections 01 33 00 and 01 33 16.00 16; for design submittals, only the final design submittal (95%) will be submitted with an AF Form 3000. Upon completion of the 95% review, the contractor will receive the submittal and AF 3000 back with one of the following three designations: Approved; Approved with Comments; or Disapproved.

- Approved: Provide the signed/sealed 'Issued for Construction' submittal in electronic format (as described below).
- Approved with Comments: Provide an unsealed 'Issued for Construction' submittal in electronic format. The electronic submittal will serve as back check set; once Government has verified that all comments have been addressed, submit signed/sealed documents (as described below).
- Disapproved: Resubmit in accordance with specification section 01 33 00.

Material submittals (SD-03, SD-04, SD-07, SD-08) for major components of the work shall be provided during the pre-Final design phase (earlier for long lead items) in accordance with 01 33 00; information provided as part of the design documents (for example, cut sheets attached to a design analysis/narrative) will not be considered as meeting submittal requirements.

Shop drawings used as design drawings shall be in accordance with the requirements of specification section 01 33 16.00 16.

3.5.3.1 Preliminary Design (35% Design)

The preliminary design shows the essential components of the proposed design and conveys that the Contractor fully understands the contract requirements and has included all elements of work to complete the project.

The documents shall include the following minimum information, as applicable to the project:

- Changes necessary to comply with Government comments/scope refinements resulting from the initial design conference.
- Site plans, floor plans, elevations, building sections, and wall sections that have been further developed from preliminary documents provided by the government.
- Single-line schematic drawings for mechanical, electrical, communications, etc.
- Listing of proposed Unified Federal Guide Specification (UFGS) sections applicable to the project.
- Design analysis that includes design narrative and analysis, including code analysis by all disciplines.
- Listing of Government furnished materials.
- Previous meeting minutes and documentation of any other communication between the Government and the Contractor that influenced the design.

3.5.3.2 Intermediate Design (65% Design)

Intermediate design builds upon the preliminary design and demonstrates that design development is in full progress. Intermediate Design can be performed in conjunction with Government approval of the Contractor's responses to the review comments on the Preliminary Design submittal.

The documents shall include the following minimum information, as applicable to the project:

- Any changes necessary to comply with Government comments to the Preliminary Design submittal.
- Complete site and landscaping plans.
- Architectural plans with complete floor and roof plans; elevations, building sections, wall sections; interior/exterior finish schedules; color boards; and other details and notes as required.
- Structural plans.
- Mechanical, fire protection, plumbing, and electrical plans with details, notes, and equipment schedules. Equipment layouts showing clearances required for maintenance and/or code.
- Specifications in rough draft (redlines/strikeouts).
- Design analysis that includes design narrative and analysis. The design analysis shall include all design assumptions and calculations for all disciplines, code analysis by all disciplines, manufacturers' material and equipment cut sheets for major equipment/fixtures.
- Updated construction schedule to include any work phasing to accommodate Government operations.

- Copies of previous design review comments and responses; previous meeting minutes; and documentation of any other communication between the Government and the Contractor that influenced the design.

3.5.3.3 Pre-Final Design (95% Design)

The pre-final design represents what the contractor believes to be a complete design, ready for construction. Pre-Final Design can be performed in conjunction with Government approval of the Contractor's responses to the review comments on the Intermediate (65%) Design.

The documents shall include the following minimum information, as applicable to the project:

- Any changes necessary to comply with Government comments to the intermediate design.
- Complete construction drawings and specifications.
- Submittal Register (generated from Contractor's specifications).
- Final design analysis.
- Material submittals (SD-03, SD-04, SD-07, SD-08) for all major components of work (generated from Contractor's specifications).
- Color boards and finishes.
- Recovered Material Determination Form (RMDF).
- Final construction schedule to include any work phasing to accommodate Government operations.
- Copies of previous design review comments and responses; previous meeting minutes; and documentation of any other communication between the Government and the Contractor that influenced the design.

3.5.3.4 Final Design (100% Design)

Final 'issued for construction' versions of documents submitted for Pre-Final (95%) review.

The documents shall include the following minimum information, as applicable to the project:

- Any changes necessary to comply with Government comments to the Pre-Final Design.
- Sealed construction drawings and specifications.
- Submittal Register (generated from Contractor's specifications).
- Sealed design analysis.
- Corrected material submittals.
- Signed RMDF.

4. DESIGN-BUILD APPROACH/OBJECTIVES

The following is provided as a starting point for the contractor's design. This information should not be considered complete; the Contractor and their Designer of Record (DOR) shall be responsible for thoroughly reviewing the information provided, performing their own site investigations and code review, and completing the design/construction for a complete and useable project.

4.1 STRUCTURAL DEMOLITION

- (a) Not anticipated.

4.2 STRUCTURAL CONSTRUCTION

4.2.1 Catwalk Netting

- (a) Install supports under the catwalk to support the netting.
- (b) Netting shall not allow tools or objects to pass through the (1/4"x1/4") openings and shall be fire retardant. Netting shall meet the following standards: OSHA, ANSI 10.37, and NFPA 701 Test method II.
- (c) Netting shall extend out past the catwalks a minimum of 3'-9" each side.
- (d) Basis of design includes clamping 10'-0" long slotted channel framing to bottom of the catwalk channels at 5'-0" on center and attaching the netting to the slotted channels. More feasible solution provided by the contractor will be considered.
- (e) Design shall include a reasonable method of debris removal from netting.
- (f) Existing structure shall be analyzed for new loads.
- (g) Grinding or disturbance of painted or flat surfaces shall be avoided. If grinding occurs on painted/flat surfaces, then testing of Asbestos/Chromium will be required prior to work.
- (h) Install warning signs to access ladders stating to "Use Caution to Prevent Falling Object Hazards to the Occupants and Equipment Below".

4.3 ORDER OF PRECEDENCE

In the event of a conflict or inconsistency between any of the provisions of this contract, precedence shall be applied in the following order:

- (1) Any portions of the accepted proposal which both conform to and exceed the requirements of the solicitation.
- (2) The provisions of the solicitation, including:
 - (a) The Solicitation, Offer and Award
 - (b) Specifications (if provided)
 - (c) Drawings
 - (d) Order SOW
- (3) All other provisions of the accepted proposal
- (4) Any contractor generated design products including, but not limited to: construction drawings, specifications, engineering studies and analyses, shop drawings, and equipment installation drawings.

4.4 RECORD DRAWINGS

4.4.1 Working As-Built Drawings ('Redline' Set)

Working as-built drawings shall be maintained at the job site, and clearly indicate authorized field changes, updates based on existing conditions, and corrections to the drawings. Reference specification section 01 78 00 for additional information related to markups to be included in working as-built drawings.

Within five (5) days of substantial completion (i.e. completion of Contractor's punch-out inspection) and prior to pre-final inspection, the Contractor shall submit working as-built drawings for Government use during the inspection. At the conclusion of the pre-final inspection, the Government and Contractor will identify any additional changes or corrections required for final approval.

4.4.2 Record Drawings

Within ten (10) days after the pre-final inspection and approval of the working as-built drawings, the Contractor shall incorporate the redlines into the CAD files and submit this final set as the Record Drawings. These drawings shall be labeled 'RECORD DRAWING AS-BUILTS' and submitted for Government approval. Reference specification section 01 78 00 for additional information.

Shop drawings shall be in accordance with the 'additional record drawings' requirements of specification section 01 78 00.

4.5 OPERATIONS & MAINTENANCE (O&M) MANUALS AND TRAINING

O&M manuals are not required for this Order.

O&M training is not required for this Order.

4.6 DD FORM 1354 REQUIREMENT

A DD Form 1354, *Transfer and Acceptance of Military Real Property*, is not required for this Order.

5. GOVERNMENT FURNISHED DRAWINGS, SPECIFICATIONS, & REFERENCES

5.1 DRAWINGS

Refer to Appendix A, DRAWING INDEX. Drawing files, in PDF and AutoCAD format, are included on the attached CD.

5.2 SPECIFICATIONS

Refer to Appendix B, SPECIFICATIONS for a for a list of administrative specification developed by the Government and applicable to the work. Specification files, in PDF format, are included on the attached CD. (Technical specifications are to be generated by the contractor as part of the design.)

5.3 BASE STANDARDS / REFERENCE DOCUMENTS / FORMS

Refer to Appendix C, TINKER AFB FACILITY STANDARDS AND REFERENCE DOCUMENTS. Tinker AFB facility standards and reference files, in PDF format, are included on the attached CD.

5.4 FACILITY AS-BUILT INFORMATION

Relevant as-built drawings for 3001 are included on the attached CD. If necessary, additional as-built drawings are available from 72 ABW/CE - Engineering Support, located in B400.

6. GOVERNMENT FURNISHED PROPERTY

There is no Government Furnished Property included in this Order.

7. STAGING/STORAGE AREA(S)

Contractor shall limit storage of materials, equipment, and incidentals to the immediate construction site while work is being performed. If the Contractor requires additional storage area(s) other than the construction site, a request shall be submitted in accordance with specification section 01 50 00, *Contractor Facilities*.

It is recommended that the Contractor arrange for storage of equipment between its delivery and its installation, as availability of space and security during non-working hours cannot be guaranteed.

Notify PMCG for staging space at least (2) days prior.

8. 'OR EQUAL' CLAUSE

Where reference is made in the specification or on the drawings to specific manufacturers and/or model numbers of equipment or materials, the information is provided to indicate design intent and to establish the minimum quality standards, physical attributes, and performance characteristics which are required for the referenced item. The use of specific names and/or model number is not intended to limit selection of other manufacturers; equipment or materials meeting these criteria may be submitted to the COR for consideration and shall be clearly identified, in writing, as a substitution.

The burden of proving equality for proposed substitutions shall be the responsibility of the Contractor and submittals for said substitutions shall include detailed manufacturer's specifications, dimensioned drawings, performance data and any other information necessary to clearly indicate that the proposed item meets or exceeds the quality standards, physical characteristics, and performance criteria established by the project documents. Submittals which do not provide proof of equality will be rejected.

Approval of a substitute by the COR shall not result in a change to Order price or schedule; such changes can only be approved by the CO.

9. ATTACHMENTS

The following project specific attachments referenced within this SOW are included on the attached CD.

- Attachment 1 – Appointment of Security Representative Letter
- Attachment 2a – Controlled Area Training
- Attachment 2b – Controlled Area Entrance Procedures
- Attachment 2c – Controlled Area Training Certificate
- Attachment 3 – [Draft] Recovered Materials Determination Form
- Attachment 4 – Project Narrative

End of SOW

APPENDIX A Drawing Index

General

G-101 General Notes and Vicinity Map

Structural

S-101 Structural Plan

APPENDIX B

Specifications

Division 00 – Procurement and Contracting Requirements

00 70 00	General Requirements for Construction on Tinker AFB
00 71 00	General Safety Requirements for Construction on Tinker AFB
00 72 00	Environmental Requirements for Construction on Tinker AFB
00 80 00	Special Provisions for Working on Tinker AFB
00 80 05	Special Provisions for Working on or Near the Airfield on Tinker AFB

Division 01 - General Requirements

01 33 00	Submittal Procedures
01 33 16.00 10	Design Data (Design After Award)
01 42 00	Sources for Reference Publications
01 45 00.00 10	Quality Control
01 50 00	Contractor Facilities
01 78 00	Closeout Submittals
01 78 23	Operation and Maintenance Data

APPENDIX C

Tinker AFB Facility Standards and Reference Documents

Base Facility Standards

Tinker Air Force Base Facility Standard
Tinker Air Force Base Mechanical Standard
Tinker Air Force Base Electrical Standard
Tinker Air Force Base Fire Protection Standard

Tinker AFB General Specifications (Div. 00 and Div. 01) Reference Documents

Sections 00 7x 00

DoDOP-2000.16V1 AFI 10-245-0	Antiterrorism (AT) Program Implementation
DAFI 13-213	Airfield Driving
DAFI 13-213 TINKERAFBSUP	Airfield Driving (Tinker AFB Supplement to AFI 13-213)
DAFI 21-101	Aircraft and Equipment Maintenance Management
DAFI 21-101 AFMCSUP	Aircraft and Equipment Maintenance Management (AFMC Supplement to AFI 21-101)
AFI 48-104	Tobacco Free Living
DAFI 91-202	The US Air Force Mishap Prevention Program
AFI 91-202 AFMCSUP	The US Air Force Mishap Prevention Program (AFMC Supplement to AFI 91-202)
AFMAN 32-7003	Environmental Conservation
AFMAN 40-201	Radioactive Material (RAM) Management
AFMAN 48-148	Ionizing Radiation Protection
DAFMAN 91-203	Air Force Occupational Safety, Fire, and Health Standards
AFMAN 91-203 OC-ALCSUP	Air Force Occupational Safety, Fire, and Health Standards (OC-ALC Supplement to AFMAN 91-203)
TINKERAFBI 13-204	Airfield Operations
TINKERAFBI 21-100	Foreign Object Damage (FOD) and Dropped Object Prevention (DOP) Programs
TINKERAFB_SOP_Outages	Planned and Emergency Outages Standard Operating Procedures
TINKERAFB HWMP	Tinker AFB Hazardous Waste Management Plan
TINKERAFB INRMP	Tinker AFB Integrated Natural Resources Management Plan
TINKERAFB INRMP_APP E	Tinker AFB Integrated Natural Resources Management Plan – Appendix E, Native Landscaping Plant Materials
TINKERAFB INRMP_APP G	Tinker AFB Integrated Natural Resources Management Plan – Appendix G, Urban Forestry Management Procedures
EM 385-1-1	Safety -- Safety and Health Requirements Manual
UFC 3-260-01	Airfield and Heliport Planning and Design
UFC 3-260-04	Airfield and Heliport Marking

APPENDIX C (Continued)

UFC 3-535-01	Visual Air Navigation Facilities
UFC 3-600-01	Fire Protection Engineering for Facilities
UFC 3-601-02	Fire Protection Systems Inspection, Testing, and Maintenance
<u>Section 00 78 00</u>	
ERDC/ITL TR-12-1	A/E/C Graphics Standard
ERDC/ITL TR-12-6	A/E/C CAD Standard
URML	Unified Master Reference List

Tinker AFB General Requirements Specifications Attachments/Forms

00 70 00, Attachment 1	Tinker AFB Outage Request Form
00 71 00, Attachment 1	Airfield Construction Waiver
00 71 00, Attachment 2	EAL Template
00 71 00, Attachment 3	VEAL Template
00 72 00, Attachment 7a	AF3952, Chemical/Hazardous Material Request Authorization Form
00 72 00, Attachment 7b	Monthly Hazardous Material Usage Report
00 72 00, Attachment 8	Recovered Materials Determination Form
00 72 00, Attachment 9	EMS Awareness Training
00 72 00, Attachment 13	AF487, Generator Operating Log
00 72 00, Attachment 14	Information for New Boilers
01 33 00, Attachment 1	TAFB Form 3000, Material Approval Submittal
00 15 00, Attachment 1	Staging Request Form
01 78 00, Attachment 1	DD1354, Transfer and Acceptance of Military Real Property
01 78 00, Attachment 2	DD1354 Checklist
SOW, Section BADGES	Appointment of Security Representative

Tinker AFB General Requirements Specifications Required Training Materials

EMS General Awareness Training for Contractors
AFMC FOD/DOP Program
Controlled Area Training (Bldg. 3001)